Starting as overdensities in the initial density fluctuations and evolving into the largest gravitationally bound structures in the Universe, galaxy clusters are a rich testbed of the cosmos. Their large masses, self-similarity, combination of both luminous and dark matter, and multiwavelength observables make galaxy clusters ideal probes of cosmology. Specifically, I am interested in using galaxy clusters to study the fundamental nature of dark energy by understanding the growth of structure in the Universe.

In order to set competitive limits, several systematic effects must be studied and mitigated, including those arising from optical cluster selection, blending-induced biases in shear and redshift calibration, and mass-estimation errors caused by model misspecification. I plan to perform early science measurements using clusters observed by Roman and Rubin while developing novel mitigation and calibration methods for future data releases. My previous experiences with clusters and shear calibration, along with membership of LSST DESC and the Rubin commissioning team equip me with the necessary tools to successfully carry out this research.

Previous and Current Research *Blending* During my PhD, I focused on the weak lensing systematic of **unrecognized blends**, which are objects (typically galaxies) that are mistakenly identified as a single entity due to severe overlap between sources (see Fig. 1 for an example). Studies of the effects of unrecognized blends on cosmology from detection bias alone have shown shifts in cosmology parameters at the level of a few percent [1–3]. An early study that also included the impact of blending on shape and flux measurements in cosmic shear found a 2σ shift in S_8 [4]. In 3x2pt cosmological analysis, expensive image simulations are used to quantify the biases unrecognized blends introduce into redshift and shear calibration. Directly identifying unrecognized blends at the catalog level is an appealing solution that I pursued during my PhD.

In [6], we showed that clustering-based **machine learning techniques can identify unrecognized blends at the catalog level**. Those techniques estimate the probability that an object is an unrecognized blend based on measured fluxes and were validated on COSMOS data. However, it was unclear whether such a removal would ultimately improve our cosmology. As mentioned earlier, previous work suggested that removing unrecognized blends from the sample could lead to tighter cosmology, but those works make assumptions that limit the applicability to modern analyses. On the other hand, the current mitigation strategy of large simulation does measure the biases present in the latest surveys but offers no explainability. To determine whether such a removal would improve cosmological constraints, I investigated the efficacy of our blend removal method on shear calibration bias using the AnaCal shear framework. We showed that our blend removal scheme **can reach LSST Year 10 shear calibration bias requirements** with a 30-40% reduction in sample size [5]. This methodology can **enable weak lensing analyses that use alternate tomographic bins** from 3x2pt analysis like in cluster lensing and dwarf galaxy lensing by offering an alternate mitigation strategy.

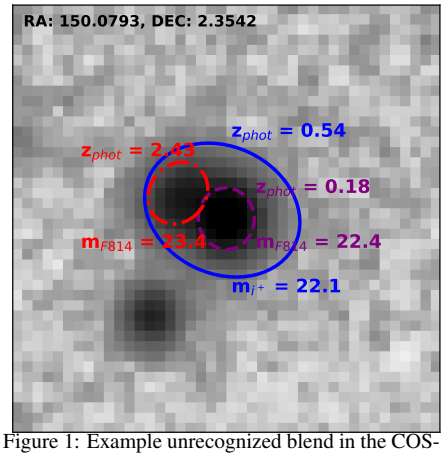


Figure 1: Example unrecognized blend in the COSMOS field with observed magnitudes and estimated photometric redshifts attached. Dark blue is the object as detected and measured by Subaru, dashed-dotted red and purple are detections and measurements from HST. From [5].

Clusters I led the LSST DESC Project 408 "A Rubin view of Abell 360", a collaborative effort between the Rubin Observatory and DESC, to study the galaxy cluster Abell 360 using commissioning data. This is the first weak lensing result to come from the Rubin Observatory and acts as proof that the instrument is able to deliver science-quality data even in suboptimal conditions. I contributed to **all steps of analysis** from red sequence identification to mass fitting. My responsibilities included **source sample selection, photo-z validation, shear measurement and calibration with shapeHSM and AnaCal, and cluster mass**

fitting. I helped collaborators create aperture mass maps, study 1D galaxy distributions, and evaluate cuts with synthetic source injection (SSI). I also led scientific discussion, coordinated paper writing efforts, and organized almost daily telecons. This was my first foray into clusters along with leading a large collaborative effort.

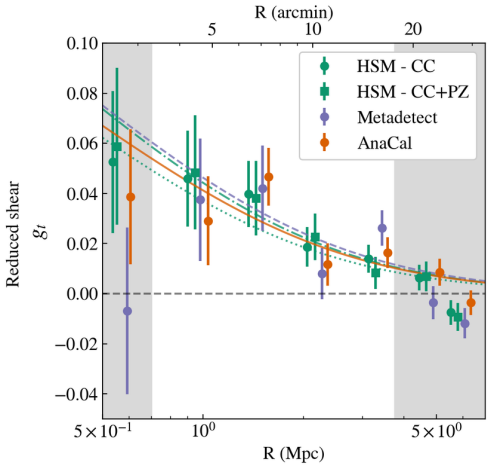


Figure 2: Tangential shear signal measured around Abell 360 using a variety of shape measurements and source selections. From [7]

Following my efforts on Abell 360, I have been testing the combination of the cutting-edge shear calibration AnaCal and photo-z estimation framework RAIL by measuring cluster profiles in the Euclid Deep Field South [8]. In this work we have **measured tangential shear signals around clusters that were undetectable in the base Rubin catalog** due to cleaner sample selection and higher quality shear estimates, highlighting the power of the combination of AnaCal and RAIL. This is the **first cluster analysis** using these tools.

Throughout my PhD, I have worked extensively on weak lensing systematics and focused on measurement validation for LSST. I have developed new weak lensing mitigation strategies [5, 6], produced some of the first weak lensing results from Rubin [7], and have a familiarity with the tools used in modern cosmological analysis [8].

Research Proposal Sub-percent level constraints on $\sigma_8(z)$ are achievable for LSST with galaxy clusters if the mass scatter can be constrained to within ±0.05 [9] whereas the latest cluster calibrations are limited to $\sim\pm0.2$ [10, 11]. The majority of that error budget comes from redshift calibration, which is impacted by blending [10–12]. Furthermore, it is **currently unknown whether modern shear calibrations, AnaCal and Metadetect, are able to meet the 1% shear calibration requirement** needed for cluster weak lensing when accounting for realistic cluster conditions [13]. Accurately characterizing biases in cluster weak lensing is critical for achieving precision cosmology. These measurements will also guide the development of specialized tools for cluster weak-lensing analyses. Over the next few years, I propose to measure the level of shear bias in the cluster regime, work on novel methods of photo-z calibration, and produce cluster lensing profiles with Roman and Rubin.

Goal 1: Shear bias Precise ensemble measurements of galaxy clusters require an estimate of both statistical and systematic bias. For clusters, this is traditionally done with N-body or hydrodynamic simulations that can capture physics-induced bias (like miscentering of optical clusters due to triaxiality), but are computationally expensive and can ignore biases coming from image processing. Directly simulating co-adds is a substantially cheaper method that has been used to calibrate biases in cosmic shear algorithms like AnaCal and Metadetect [14–17]. However, the applicability of this calibration to the cluster regime — which experiences higher shear and higher rates of blending — has not been established.

Accurately quantifying the shear calibration bias will involve incorporating: foreground cluster members and contaminants, intra-cluster light, cluster triaxiality, and more realistic sky background variation. This will then be compared to the calibration bias measured by the standard image simulation technique. Adding these systematics sequentially will allow us to isolate the bias introduced by each effect and inform future mitigation strategies.

Goal 2: Photo-z calibration and joint processing As stated above, photo-z calibration, which is affected by blending and unrecognized blends, remains a significant source of systematic error for cluster cosmology. My previous work on removing contaminated objects is a straightforward method that may improve the photo-z calibration at a statistical cost. An alternative approach is to use the machine learning probability

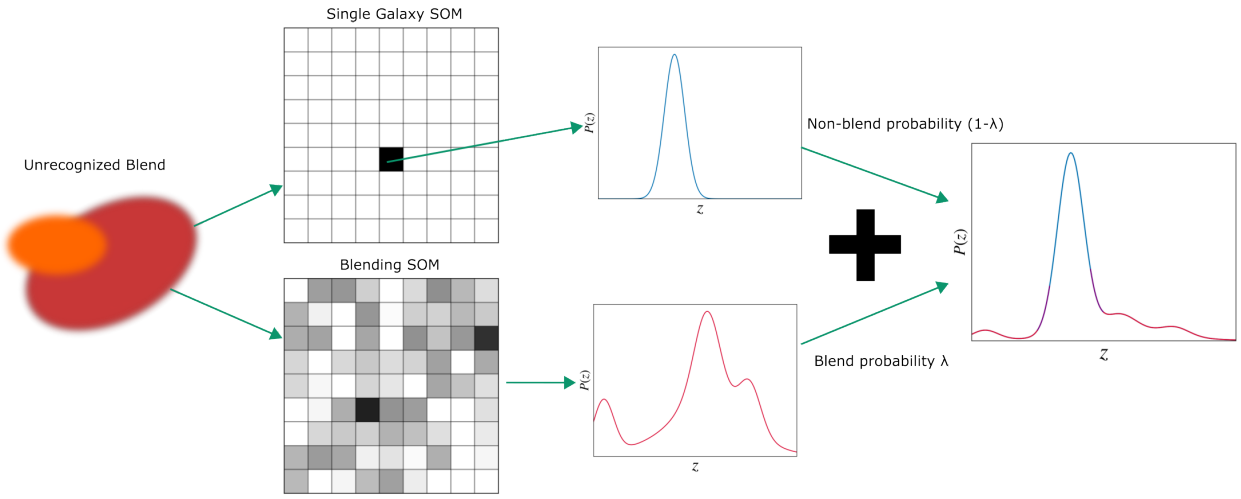


Figure 3: Example of modeling blends when producing redshift estimates via a SOM. Instead of only using the cell of the best fit, a weighted SOM can be made from synthetic blends that can be used to produce a more accurate PDF (color-coded to highlight the contribution from individual terms).

from [6] as a prior and model certain objects as a superposition of two galaxies. This may lead to more accurate posteriors for redshift without incurring a statistical cost.

The self-organizing map (SOM) is a powerful machine learning algorithm that projects the multidimensional color space onto a 2D surface. The same map that is used to create ensemble redshift estimates [18] can be extended to account for blends as outlined in Fig. 3. In that example, we use a blending probability (the prior from [6]) to combine two redshift probability distributions: the distribution assuming a single galaxy and the distribution assuming a superposition of two galaxies. In preliminary work, I have shown that this approach can produce more accurate posterior estimates than a single-galaxy SOM approach. Accounting for more realistic blending, training-testing mismatch, and sample size variance are all necessary before applying this approach to real data.

The most effective solution for unrecognized blends would be to recognize them through joint processing of space- and ground-based imaging. While there are strong solutions on how to best combine photometry, there is work to be done on combining shape measurements in order to maximize weak lensing signal. During my postdoc, I plan to contribute to the ongoing efforts to enable joint-processing for both Metadetect and AnaCal. This will mostly rely on simulated datasets, like the existing Roman-Rubin simulations, but can be tested on data using overlapping deep-field observations of clusters which I am well equipped to study. If successful, this will completely circumvent the need for the machine learning techniques I have previously worked on wherever there is overlap between ground- and space-based instruments. Regardless of the solution used, improving the photo-z calibration and reducing shape noise at higher redshifts are key to constraining the growth of structure with clusters.

Goal 3: Cluster measurements for Roman and Rubin With a validated pipeline in hand, we will have confident (or at least properly weighted) cluster weak lensing masses. Rubin Data Preview 2 includes ~ 100 X-ray detected clusters by eRosita and ~ 400 SZ detected clusters by ACT. Each sample is large enough to produce an LSST mass calibration, and given that the previous ACT calibration uses a similar number of clusters [19], it is possible to produce competitive cosmological results using this early dataset. Constraining cosmology from calibration constitutes another project that would require estimating a covariance, modeling shear profiles (either hierarchically or stacked), redshift distribution estimation, and running cosmology chains. As an active member in LSST DESC, I am excited to contribute my skills to these efforts and help build a tomographic understanding of the cosmos.

I will continue working with galaxy clusters and use them to set competitive constraints on cosmology. I am uniquely positioned to measure and mitigate weak lensing systematics relevant to cluster cosmology. Understanding and accounting for these biases is key to probing the Universe tomographically.

References

[1] B. Levine et al., "Galaxy Clustering with LSST: Effects of Number Count Bias from Blending", The Open Journal of Astrophysics **8**, 38, 38 (2025).

[2] J. Hartlap et al., "A bias in cosmic shear from galaxy selection: results from ray-tracing simulations", Astronomy and Astrophysics **528**, A51, A51 (2011).

[3] E. Genc et al., "Quantifying the impact of detection bias from blended galaxies on cosmic shear surveys", arXiv e-prints, arXiv:2507.01546, arXiv:2507.01546 (2025).

[4] E. Nourbakhsh et al., "Galaxy blending effects in deep imaging cosmic shear probes of cosmology", Monthly Notice of the Royal Astronomical Society **514**, 5905–5926 (2022).

[5] P. Adari et al., "Improvements to Shear Calibration Bias via Removal of Blends from Different Redshifts", (in prep.).

[6] S. Liang et al., "Catalog-based detection of unrecognized blends in deep optical ground based catalogs", arXiv e-prints, arXiv:2503.16680, arXiv:2503.16680 (2025).

[7] A. von der Linden et al., "A Weak-Lensing View of Abell 360 in Vera Rubin Observatory's Data Preview 1", (in prep.).

[8] P. Adari et al., "Cluster lensing measurements from Vera Rubin Observatory's Data Preview 1 using AnaCal and RAIL", (in prep.).

[9] H.-Y. Wu et al., "Cosmology with Galaxy Cluster Weak Lensing: Statistical Limits and Experimental Design", Astrophysical Journal **910**, 28, 28 (2021).

[10] S. Bocquet et al., "SPT clusters with DES and HST weak lensing. I. Cluster lensing and Bayesian population modeling of multiwavelength cluster datasets", Physical Review D. **110**, 083509, 083509 (2024).

[11] S. Grandis et al., "The SRG/eROSITA All-Sky Survey: Dark Energy Survey year 3 weak gravitational lensing by eRASS1 selected galaxy clusters", Astronomy & Astrophysics **687**, A178, A178 (2024).

[12] F. Köhlinger et al., "Statistical uncertainties and systematic errors in weak lensing mass estimates of galaxy clusters", Monthly Notices of the Royal Astronomical Society **453**, 3107–3119 (2015).

[13] The LSST Dark Energy Science Collaboration et al., "The LSST Dark Energy Science Collaboration (DESC) Science Requirements Document", arXiv e-prints, arXiv:1809.01669, arXiv:1809.01669 (2018).

[14] N. MacCrann et al., "Dark Energy Survey Y3 results: blending shear and redshift biases in image simulations", Monthly Notice of the Royal Astronomical Society **509**, 3371–3394 (2022).

[15] S. Mau et al., "Dark Energy Survey Year 6 Results: Image Simulation for Weak Lensing Shear and Photometric Redshift Calibration", (in prep.).

[16] X. Li and R. Mandelbaum, "Analytical weak-lensing shear responses of galaxy properties and galaxy detection", Monthly Notice of the Royal Astronomical Society **521**, 4904–4926 (2023).

[17] E. S. Sheldon et al., "Metadetection Weak Lensing for the Vera C. Rubin Observatory", The Open Journal of Astrophysics **6**, 17, 17 (2023).

[18] J. Myles et al., "Dark Energy Survey Year 3 results: redshift calibration of the weak lensing source galaxies", Monthly Notice of the Royal Astronomical Society **505**, 4249–4277 (2021).

[19] T. Shin et al., "Weak Lensing Mass Calibration of the ACT DR5 Galaxy Clusters with the DES Year 3 Weak Lensing Data", Journal of Cosmology and Astroparticle Physics **2026**, 075, 075 (2026).